

Gosford High School

Year 11

2008

Preliminary Higher School Certificate

Mathematics

Assessment Task 2

Time Allowed - 60 minutes
(+ 5 minutes Reading Time)

Remember to start each new question on a new page

Students must answer questions using a blue/black pen and/or a sharpened B or HB pencil.

Approved scientific calculators may be used

Students need to be aware that

- * 'bald' answers may not gain full marks.
- * untidy and/or poorly organised solutions may not gain full marks.

Question 1 (15 marks)

- (a) Evaluate $|-5| - |-8|$ (1)
- (b) Write $\frac{7}{11}$ correct to 2 decimal places (1)
- (c) Find $\frac{\sqrt{573 \cdot 6 + 114 \cdot 9}}{3\pi}$ correct to the nearest whole number (1)
- (d) In 2007 Council rates increased by $7\frac{1}{2}\%$. The new rate for a property is \$1735. What was the old rate for this property? (answer to the nearest dollar) (2)
- (e) Write 0.0384 correct to 2 significant figures (1)
- (f) Simplify $3\sqrt{32} - 6\sqrt{18} + 2\sqrt{24}$ (2)
- (g) Find a if $\frac{4\sqrt{2} \times \sqrt{12}}{2\sqrt{6} \times 3\sqrt{8}} = a\sqrt{2}$ (3)
- (h) If $a = \sqrt{2} - 1$ find the exact value of $a + a^{-1}$ (3)

Question 2 (12 marks)

(a) Factorise each of the following

(i) $10a^2 + 3a - 4$ (1)

(ii) $8x^3 + 512$ (1)

(ii) $x(3x - 7) - (3x - 7)^2$ (2)

(iii) $p^2 - q^2 - p + q$ (2)

(b) Expand and simplify

(i) $(3y - 4x)^2$ (1)

(ii) $(2m - 1)(4m^2 + 2m + 1)$ (1)

(c) Simplify $\frac{6}{x^2 - 9} - \frac{x - 6}{x^2 - 3x - 18}$ (3)

Question 3 (12 marks)

(a) Solve

(i) $6x^2 = 9x$ (2)

(ii) $2\left(\frac{1}{a} + 4\right) = 5 - \frac{2}{a}$ (2)

(iii) $|4x - 11| = 5x + 2$ (3)

(iv) $x = \sqrt{3x + 4}$ (3)

(b) Solve $2t^2 = 3t + 8$ writing your answer in surd form. (2)

Question 4 (12 marks)

(a) Graph $\{x: -1 < x \leq 2\}$ on a number line. (1)

(b) Solve

(i) $(2x + 3)(x - 4) \geq 0$ (2)

(ii) $|4x - 3| > 13$ (2)

(iii) $\frac{x+2}{-3} \geq \frac{x-5}{2}$ (2)

(c) Solve the following pairs of simultaneous equations

(i)
$$\begin{aligned} 2c - 3d &= 4 \\ 3c + d &= 17 \end{aligned}$$
 (2)

(ii)
$$\begin{aligned} xy &= -4 \\ 2x + y &= 2 \end{aligned}$$
 (3)

Question 5 (9 marks)

(a) The equation $x^2 - 2ax - a = 0$ has one root at $x = -5$. Find the value of a . (1)

(b) Solve $27^x = \frac{1}{81}$ (2)

(c) Show that the equation $2x^2 - 5x + 8 = 0$ has no real solutions (2)

(d) Use the formula $S = \frac{n}{2}[2a + (n-1)d]$ to find the value of d when $n = 9, a = -6$ and $S = 540$ (2)

(e) Use the method of 'Completion of the Square' to solve the equation $x^2 - 8x + 9 = 0$ (2)

Solutions to Assessment 2

$$\therefore a) |-5| - |-8| = 5 - 8 \\ = -3$$

$$b) \frac{7}{11} = 0.6363... \\ = 0.64 \text{ to 2 d.p.}$$

$$c) \frac{\sqrt{573.6 + 114.9}}{(3\pi)} = 2.784... \\ = 3 \text{ to the nearest whole number}$$

$$d) 107\frac{1}{2}\% \text{ of Old Rate} = \text{New Rate} \\ 1.075 \times \text{Old Rate} = \$1735 \\ \text{Old Rate} = \frac{\$1735}{1.075} \\ = \$1613.953... \\ = \$1614 \text{ (to the nearest dollar)}$$

$$e) 0.0384 = 0.038 \text{ (to 2 sign. figures)}$$

$$f) 3\sqrt{32} - 6\sqrt{18} + 2\sqrt{24} = 12\sqrt{2} - 18\sqrt{2} + 4\sqrt{6} \\ = 4\sqrt{6} - 6\sqrt{2}$$

$$g) \frac{4\sqrt{2} \times \sqrt{12}}{2\sqrt{6} \times 3\sqrt{8}} = \frac{4\sqrt{24}}{6\sqrt{48}} \\ = \frac{2}{3\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$$

$$(iii) x(3x-7) - (3x-7)^2 = (3x-7)(x - (3x-7)) \\ = (3x-7)(x - 3x + 7) \\ = (3x-7)(7-2x)$$

$$(iv) p^2 - q^2 - p + q = (p-q)(p+q) - 1(p-q) \\ = (p-q)((p+q) - 1) \\ = (p-q)(p+q-1)$$

$$b) i) (3y - 4x)^2 = 9y^2 - 24xy + 16x^2$$

$$ii) (2m-1)(4m^2+2m+1) = 8m^3 + 4m^2 + 2m - 4m^2 - 2m - 1 \\ = 8m^3 - 1$$

$$c) \frac{6}{x^2-9} - \frac{x-6}{x^2-3x-18} = \frac{6}{(x-3)(x+3)} - \frac{x-6}{(x-6)(x+3)} \\ = \frac{6}{(x-3)(x+3)} - \frac{1}{x+3} \\ = \frac{6 - 1(x-3)}{(x-3)(x+3)} \\ = \frac{9-x}{(x-3)(x+3)}$$

$$= \frac{1}{3}\sqrt{2}$$

$$\therefore a = \frac{1}{3}$$

$$h) a + a^{-1} = a + \frac{1}{a}$$

$$= \sqrt{2} - 1 + \frac{1}{\sqrt{2} - 1}$$

$$= \sqrt{2} - 1 + \frac{1}{\sqrt{2} - 1} \times \frac{\sqrt{2} + 1}{\sqrt{2} + 1}$$

$$= \sqrt{2} - 1 + \frac{\sqrt{2} + 1}{1}$$

$$= \sqrt{2} - 1 + \sqrt{2} + 1$$

$$= 2\sqrt{2}$$

$$2. a) i) 10a^2 + 3a - 4 = (5a + 4)(2a - 1)$$

$$ii) 8x^3 + 512 = 8(x^3 + 64) \\ = 8(x+4)(x^2 - 4x + 16)$$

$$3. a) (i) 6x^2 = 9x \\ 6x^2 - 9x = 0 \\ 3x(2x-3) = 0 \\ x = 0, \frac{3}{2}$$

$$(ii) 2\left(\frac{1}{a} + 4\right) = 5 - \frac{2}{a} \\ \frac{2}{a} + 8 = 5 - \frac{2}{a} \\ \frac{4}{a} = -3$$

$$4 = -3a$$

$$a = -\frac{4}{3}$$

$$(iii) |4x-11| = 5x+2$$

$$\therefore 4x-11 = 5x+2 \text{ or } 4x-11 = -(5x+2) \\ -13 = x \quad 4x-11 = -5x-2$$

$$\text{Test } |-63| \neq -63$$

Not a solution

$$9x = 9$$

$$x = 1$$

$$\text{Test } |-7| = 7$$

Is a solution

$$\therefore x = 1 \text{ only}$$

3. (iv) $x = \sqrt{3x+4}$

$$x^2 = 3x+4$$

$$x^2 - 3x - 4 = 0$$

$$(x-4)(x+1) = 0$$

$$\therefore x = 4, -1$$

Test $x = 4$

Test $x = -1$

$$4 = \sqrt{16}$$

$$-1 = \sqrt{1}$$

Is a solution

Is not a solution

$x = 4$ is the only solution

b) $2t^2 - 3t - 8 = 0$

$$t = \frac{3 \pm \sqrt{(-3)^2 - 4(2)(-8)}}{4}$$

$$t = \frac{3 \pm \sqrt{9 + 64}}{4}$$

$$t = \frac{3 \pm \sqrt{73}}{4}$$

(iii) $\frac{x+2}{-3} \geq \frac{x-5}{2}$

Multiply both sides by (-6)

$$2(x+2) \leq -3(x-5)$$

$$2x+4 \leq -3x+15$$

$$5x \leq 11$$

$$x \leq 2\frac{1}{5}$$

c) i) $3c + d = 17$

$$d = 17 - 3c \dots (1)$$

$$2c - 3d = 4 \dots (2)$$

subst. (1) into (2)

$$2c - 3(17 - 3c) = 4$$

$$2c - 51 + 9c = 4$$

$$11c = 55$$

$$c = 5$$

$$\therefore d = 17 - 3(5)$$

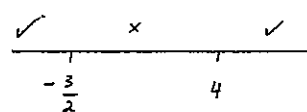
$$d = 2$$

$$\therefore c = 5, d = 2$$

b) i) $(2x+3)(x-4) \geq 0$

Solve $(2x+3)(x-4) = 0$

$$\therefore x = -\frac{3}{2}, 4$$



Test $x = -2$; $6 \geq 0$ True

$x = 0$; $-12 \geq 0$ False

$x = 5$; $13 \geq 0$ True

$$\therefore x \leq -\frac{3}{2} \text{ or } x \geq 4$$

ii) $|4x-3| > 13$

$$\therefore 4x-3 > 13 \text{ or } 4x-3 < -13$$

$$4x > 16$$

$$4x < -10$$

$$x > 4$$

$$x < -2\frac{1}{2}$$

Test $x = 5$, $|17| > 13$ True

$x = -3$, $|-15| > 13$ True

$$\therefore x < -2\frac{1}{2} \text{ or } x > 4$$

4 c) (ii) $y = -\frac{1}{x}$

$$2x + y = 2 \dots (2)$$

Subst. (1) into (2)

$$\therefore 2x + \left(-\frac{1}{x}\right) = 2$$

$$2x^2 - 4 = 2x$$

$$2x^2 - 2x - 4 = 0$$

$$x^2 - x - 2 = 0$$

$$(x-2)(x+1) = 0$$

$$\therefore x = 2, -1$$

when $x = 2$, $y = -\frac{1}{2}$

when $x = -1$, $y = 1$

5. a) $x = -5$ satisfies

$$\therefore (-5)^2 - 2a(-5) - a = 0$$

$$25 + 10a - a = 0$$

$$\therefore 9a = -25$$

$$a = \frac{-25}{9}$$

$$a = -2\frac{7}{9}$$

$$b) 27 = \frac{1}{81}$$

$$(3^3)^x = 3^{-4}$$

$$3^{3x} = 3^{-4}$$

$$3x = -4$$

$$x = -\frac{4}{3}$$

$$\begin{aligned} c) \quad b^2 - 4ac &= (-5)^2 - 4(2)(8) \\ &= 25 - 64 \\ &= -39 < 0 \end{aligned}$$

$\therefore$ Equation has no real solutions

$$d) \quad S = \frac{n}{2} [2a + (n-1)d]$$

$$540 = \frac{9}{2} [2(-6) + 8d]$$

$$540 = \frac{9}{2} [8d - 12]$$

$$540 = 9(4d - 6)$$

$$60 = 4d - 6$$

$$4d = 66$$

$$d = 16\frac{1}{2}$$

$$e) \quad x^2 - 8x + 16 = -9$$

$$x^2 - 8x + 16 = -9 + 16$$

$$x^2 - 8x + 16 = -9 + 16$$

$$(x-4)^2 = 7$$

$$x - 4 = \pm \sqrt{7}$$

$$x = 4 \pm \sqrt{7}$$